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Silent Expressions Unveiled: Deep Learning for British and American Sign Language Detection

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Abstract

Sign language is an efficient means of communication for the hearing impaired. The main goal of this research is to recognize both American and British sign languages using computer vision algorithms and neural networks. Our work addresses the critical need for inclusive and accessible communication for the Deaf community. For the study, we compiled a large dataset of her ASL and BSL gestures. Preprocess the data to improve feature extraction and reduce noise. Our model architecture uses a convolutional neural network (CNN) to capture the temporal patterns of finger gestures. This allows you to use Mediapipe keypoint detection within your model. The end result is a comprehensive model that can accurately and quickly recognize and classify sign language gestures. We further improve the usability of the system by integrating keyword recognition into the model, allowing sign language sentences to be decoded and translated into text. Our test results demonstrate how well our deep learning strategy works to achieve high efficiency and accuracy in both ASL and BSL detection. This program helps bridge the gap between those who use sign language and those who do not. Our new approach not only accurately recognizes American and British sign language, but also seamlessly integrates real-time feedback to help users make immediate improvements to improve their sign language skills. We can do better. Moreover, our approach is flexible enough to be used in a variety of settings, ensuring its usefulness in educational environments and providing sign language learners with a fun and dynamic learning environment. Explore and exploit the full potential of Mediapipe Keypoint using CNNs and also leverage the efficiency of computer vision.

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Keywords: Sign language recognition, BSL, ASL, deep learning, neural networks, computer vision, gesture recognition, assistive technology, communication accessibility, keyword detection , Convolutional Neural Network(CNN)

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1. Introduction

When it comes to helping people with hearing problems communicate, sign language is essential. Through the perspective of deep learning, the study sets out on a transformational journey to examine the domains of American and British Sign Language. Sign languages are distinctive and potent modes of expression in a world where languages and communication methods are varied. The Deaf and hard-of-hearing communities rely heavily on these languages, but technology that can automatically detect and understand sign language motions is still desperately needed.

The project's motivation arises from a sincere desire to achieve accessibility and inclusion. We are aware of the significant influence that clear communication may have on the lives of those who identify as deaf or hard of hearing. Rich languages like BSL and ASL, each with its own syntax and lexicon, are essential for fostering interpersonal relationships and exchanging ideas.The combination of deep learning and computer vision technology has opened up new ways to close communication gaps in the modern digital world. By combining artificial intelligence with the complex intricacies of sign languages, the research hopes to provide people with the tools to recognise, interpret, and comprehend BSL and ASL with ease and accuracy.

The main goal of this research is to create a deep learning-based program that will able to identify and recognise signals in both ASL and BSL. Using computer vision methods and neural network power[1], our goal is to develop a dependable and effective tool that can recognise and interpret sign language motions in real-time. The study's goal is to enhance communication accessibility for people with hearing impairments and progress assistive technology in the process.The first step of the research is gathering a large dataset of BSL and ASL gestures from variety of signers, backgrounds, and lighting situations. The deep learning models are trained and tested using this dataset as the basis. Our preprocessing of the data reduces noise and improves feature extraction, so the models can function reliably in a variety of settings.Convolutional Neural Networks are used in the suggested method for image-based sign identification. Mediapipe keypoint detection may also be used inside the model architecture to potentially capture temporal patterns in finger motions.

To further improve the system's usability, the study incorporates keyword recognition into the model to translate signed sentences into text[2].By disseminating this research, the team hopes to advance the area of assistive technology and promote inclusion by giving the Deaf and hard-of-hearing groups a useful communication tool. Primary goal of this study is to show that deep learning can be used to close communication gaps and improve accessibility so that everyone, regardless of hearing ability, may benefit from it.

Furthermore, our study explores the social and cultural dimensions of communication in sign language, acknowledging the significance of maintaining linguistic variety in Deaf and hard-of-hearing societies. Our initiative attempts to develop a technical answer, but it also wants to foster a better appreciation for these complex languages by showcasing the distinctive grammar, syntax, and regional variants found in both American Sign Language (ASL) and British Sign Language (BSL). By taking an inclusive stance, we hope to build a harmonic link between the digital age and the complex web of sign languages, encouraging tolerance and understanding between people with various hearing and language backgrounds.

2. Literature Review

Using hand, body, and face expressions, C. Raghavachari et al [3] primarily concentrated on the community of hearing and speech-challenged people. Effective translation, however, requires skilled interpreters in both directions, which results in social exclusion. Convolutional Neural Networks (CNN) have been used to construct a vision-based finger typing system that outperforms ResNet and VGG16 models while maintaining a low accuracy, performing similarly to InceptionV3.A percentage.

Study conducted by A. M. Arjun et al [4] focused on demonstrating that: (1) non-verbal hand gestures may communicate meanings without the need for face emotions; and (2) when combined, they can produce new meanings. We developed an image classification system using Convolutional Neural Networks, TensorFlow, and OpenCV. It is confined to modest size datset and is trained to identify emotions from hand movements without face expressions.A percentage.

The work that Sundar B et al do is related to deaf-mute communication, home automation, robot control, HCI, and healthcare. This article describes a vision-based method that uses Google's Mediapipe for hand landmarks to identify

American Sign Language (ASL) letter movements. With all 26 alphabets, LSTM achieves 99 percent accuracy in gesture recognition. For more effective communication, this technology translates hand motions into text; nonetheless, it would be interesting to view the translated messages in real time. [5].

The deaf population benefits from the research of L. VB, S. KB, P. H, and S. Abhishek. Using deep learning, the research translates ASL movements into text or voice. Through Using hand motion data, compare the models of VGG16, ResNet, and AlexNet. Outcomes: VGG16 achieves 93.2 percent accuracy, ResNet achieves 96.9 percent, and AlexNet achieves 95.87 percent accuracy; nevertheless, this can occasionally result in undefitting.[6]Simonyan etal paper focuses mainly on a novel method of picture categorization using deep convolutional neural networks. Nevertheless, the study's evaluation was restricted to a small section of the ImageNet dataset, and it neglected to analyse the dataset's low diversity or evaluate performance on other datasets.[7]

The main focus of Aloysius et al work is a vision-based continuous sign language recognition system. Their proposed methodology mostly relies on hand detection and tracking phases, which could not operate consistently in different contexts. Furthermore, using a KNN classifier may not be the best or most accurate option.[8]

3. Dataset

We built this data set with the intention of eliminating mistakes and providing greater diversity in the alternatives available for the model's training phase. The data-set's goal is to forecast a user's sign language so that we can decipher the message they intended to convey. These examples were chosen from a bigger database under a number of restrictions. The files include one target picture and many low light images. Fig 1,Fig 2,Fig 3 depict the sample images oof A,B,C. There is clear representation of hand poses of american and british sign with its corresponding symbol.

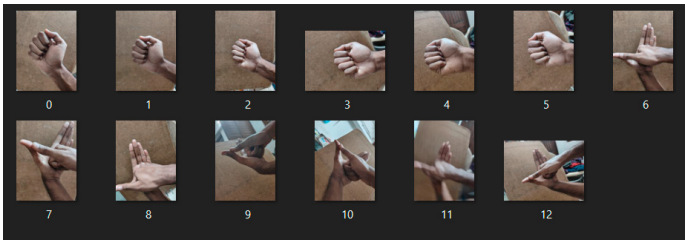


Fig. 1. Sample images from dataset of Letter A

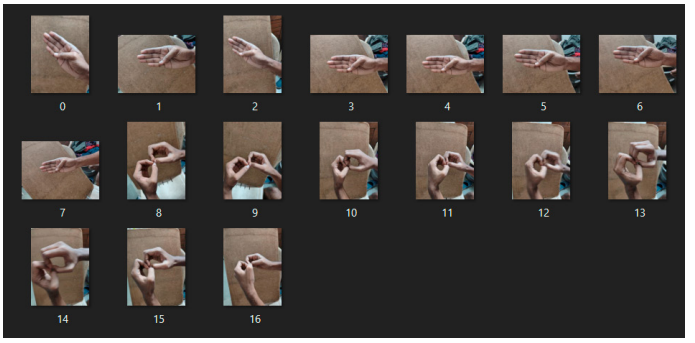


Fig. 2. Sample images from dataset of Letter B

4. Methodology

It is a difficult but worthwhile undertaking to use deep learning to detect and discriminate between American and British Sign Language in order to promote effective communication accessibility.

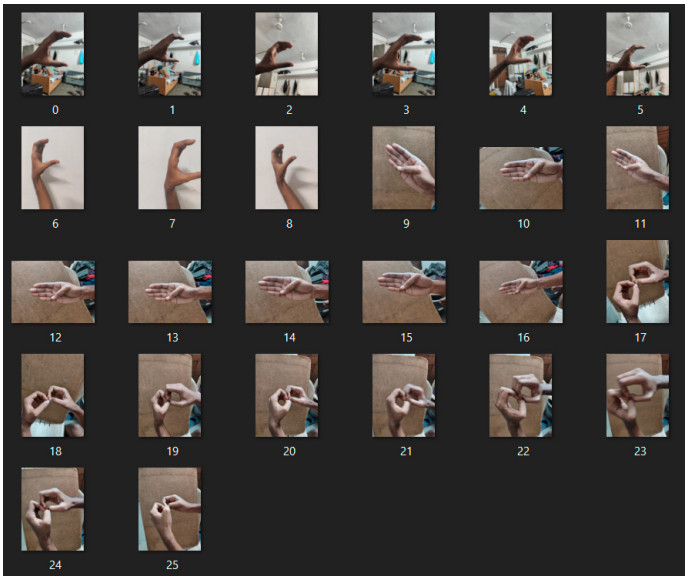


Fig. 3. Sample images from dataset of Letter C

4.1. Image Acquisition

The media Pipe library processes the input initially for hand posture estimation. Typically, the input comprises of video frames or photos with a hand motion. Key hand points, such as finger joints and locations, are detected and estimated by MediaPipe's deep learning algorithms.

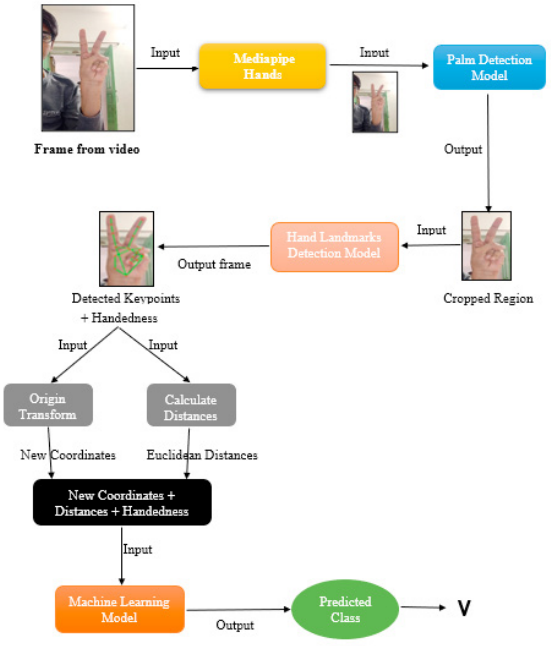


Fig. 4. Methodology

4.2. Preprocessing

Preprocessing procedures may be applied to the input data, whether it is for hand sign recognition or another purpose. To prepare the data for additional processing, this may entail shrinking photos, normalising pixel values, and converting them to grayscale.

4.3. Hand Pose Estimation

The media Pipe library processes the input initially for hand posture estimation. Typically, the input comprises of video frames or photos with a hand motion. Key hand points, such as finger joints and locations, are detected and estimated by MediaPipe's deep learning algorithms.

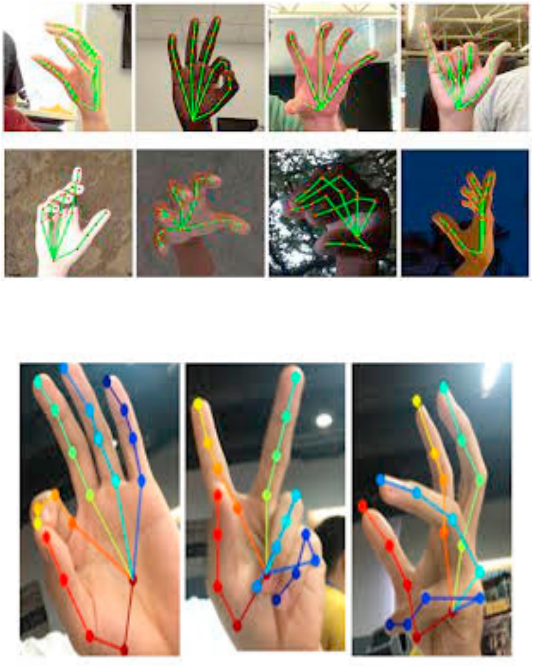


Fig. 5. Keypoint detection

4.4. Data Collection for Finger Gesture Recognition

The system gathers information about finger motions if the input is meant to be used for finger gesture recognition. Key information from MediaPipe as well as finger movement temporal patterns may be included in this data.

4.5. Finger Gesture Recognition Model

A different machine learning model—possibly utilising LSTM networks—may be used to recognise finger gestures. This model produces classification results after being trained to identify temporal patterns in finger motions.

4.6. Model Output

The model(s) generate an output based on the particular job (finger gesture recognition, hand sign recognition, or both). The output can be displayed or utilised for additional processing or interaction inside an application or system. It can contain the recognised hand sign or finger gesture and the kind of sign language employed, such as American or British Sign Language .

4.7. Confusion Matrix

A confusion matrix gives an idea of the performance of the working flow of the system. It shows the number of predictions that are predicted correctly and the predictions that are predicted incorrectly. It play keyrole in measuring the accuray,precision and recall.

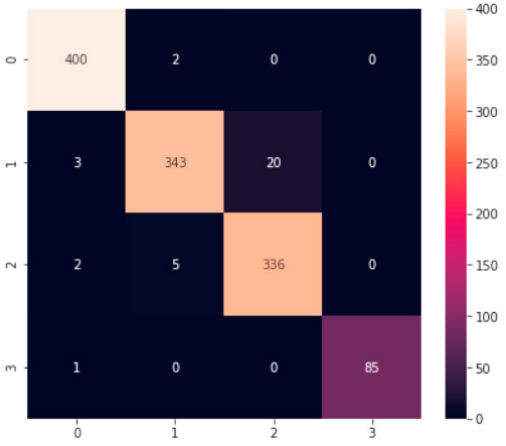


Fig. 6. Confusion matrix with 1200 samples

Classification Report

	precision	recall	f1-score	support
0	0.99	1.00	0.99	402
1	0.98	0.94	0.96	366
2	0.94	0.98	0.96	343
3	1.00	0.99	0.99	86
accuracy			0.97	1197
macro avg	0.98	0.98	0.98	1197
weighted avg	0.97	0.97	0.97	1197

Fig. 7. Classification Report

4.8. Customization and Extension

Users may be able to upload more training data or change the models to identify new hand signals or finger motions depending on the input given by customising and extending the system.

In summary when the project is implemented the system takes input throught camera and convert the video into frames and the frames are then used by the model and the key point are detected in the frames which then are compared and the symbol is detected.

5. Proposed Models

5.1. Mediapipe Keypoint Detection using CNN

One way to use MediaPipe and CNNs for sign language detection is MediaPipe Keypoint Detection using CNN for Sign Language Detection. An open-source platform for creating multimodal pipelines in real time is called MediaPipe. For applications like position estimation, hand tracking, and facial identification, it offers several pre-trained models.One kind of deep learning model that works well for image identification applications is CNNs. They may

be applied to the extraction of visual characteristics, including colours, forms, and edges. The photos may then be categorised using these attributes.[9]

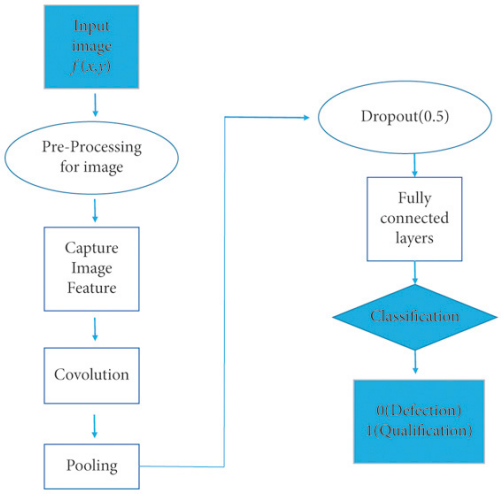


Fig. 8. Flowchart of CNN

The following actions are usually taken in order to use MediaPipe Keypoint Detection with CNN for sign language detection:

1. To find the hand keypoints in the video frames, MediaPipe is utilised.

2. From the video frames, the hand keypoints are collected and sent to CNN.

3. The hand keypoints are categorised by CNN as a specific sign language motion.

Comparing this method to more conventional sign language identification techniques, there are several advantages. It is more accurate and noise-resistant, to start. Second, it has real-time sign language gesture detection capabilities. Thirdly, it can recognise motions in sign language from many perspectives.[10]

5.2. RNN using LSTM

An RNN type called long short-term memory (LSTM) was created especially to manage long-term dependencies in sequential data. Compared to conventional RNNs, LSTMs have a number of advantages, including as the capacity to learn long-term dependencies and greater noise resistance.[11]
Because RNNs employ LSTMs to learn the temporal patterns of hand motions, they may be employed for detection of sign language. The RNN is trained on a dataset of movies showing sign language, each video labelled with the associated sign language word or phrase, in order to do this.
The RNN is trained to identify specific sign language words or phrases in the films by extracting characteristics from them, such as hand stance and movement.[12]
There are several benefits that RNNs employing LSTMs offer over conventional sign language detection techniques. Initially, they might pick up long-term dependencies in motions used in sign language.
This is significant because a lot of hand motions used in sign language require a certain order of motion. Second, compared to conventional sign language identification techniques, RNNs employing LSTMs are more noise-resistant. This is crucial because, depending on a variety of variables, including occlusion and illumination, sign language motions can be loud.[13]

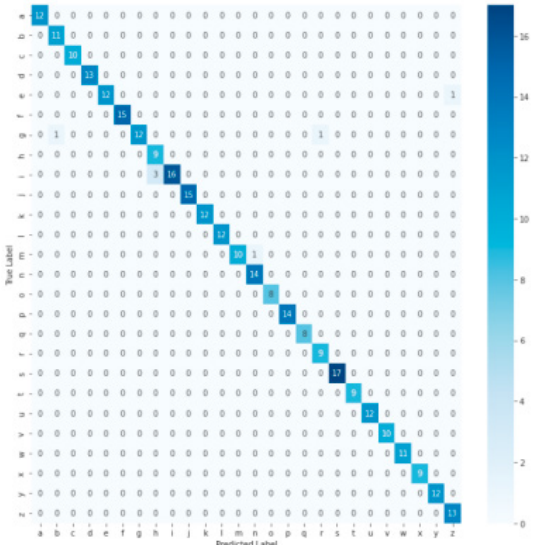


Fig. 9. Confusion matrix for sign language datase using CNN

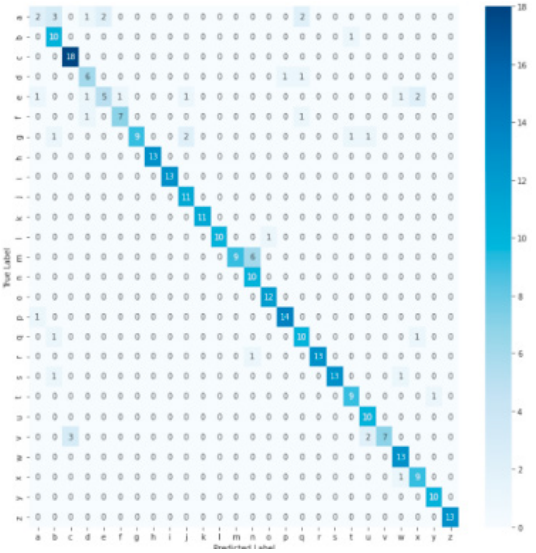


Fig. 10. Confusion matrix for sign language datase using LSTM

5.3. Dynamic Time Warping (DTW)

A technique for combining DTW with K-nearest neighbours (KNN) for sign language identification is called Dynamic Time Warping (DTW) KNN.

Even if two time series have different durations and time scales, they can still be aligned using the DTW method, which measures their similarity. KNN is a classification technique that uses the classes of its K most similar neighbours to predict the class of a new data point.[14]

In order to use DTW KNN for sign language detection, the following procedures are usually followed:

1. Sign language motions are compiled into a dataset. A temporal series of hand keypoints is used to represent each gesture in the dataset.

2. Every gesture in the dataset and the gesture that has to be recognised are separated by the DTW distance.

3. The gesture to be recognised and the K most comparable motions are discovered.

4. Based on the classes of the K most comparable motions, the class of the gesture to be detected is predicted.

Comparing this method to more conventional sign language identification techniques, there are several advantages. First off, it can withstand noise better and changes in the timing and pace of the motions.[16]

6. Results and Discussions

We have applied 3 different machine learning algorithms namely Convolutional neural network(CNN), Recurrent neural network(RNN) and Dynamic Time Warping (DTW) on the same dataset calculated the accuracy of the model by training and testing the dataset and for this we have pre-processed the data and the calculated the accuracy and the results were listed in the table below

Model	Accuracy(%)
mediapipe keypoint detection using CNN	97.24
RNN using LSTM	87.4
DTW-KNN	83.7

Our findings demonstrated that LSTM models were more effective at recognizing gesture sign language gestures, whereas CNN models were more effective at recognizing static sign language gestures and after testing these three models the conclusion is that mediapipe keypoint detection using CNN achieved an accuracy of 97.24 percenttage while RN using LSTM achieved 87.4 and DTW-KNN got 83.7 percentage accuracy showcasing the efficiency of mediapipe keypoint detection using CNN in sign language detection using the help of Open-CV in live.

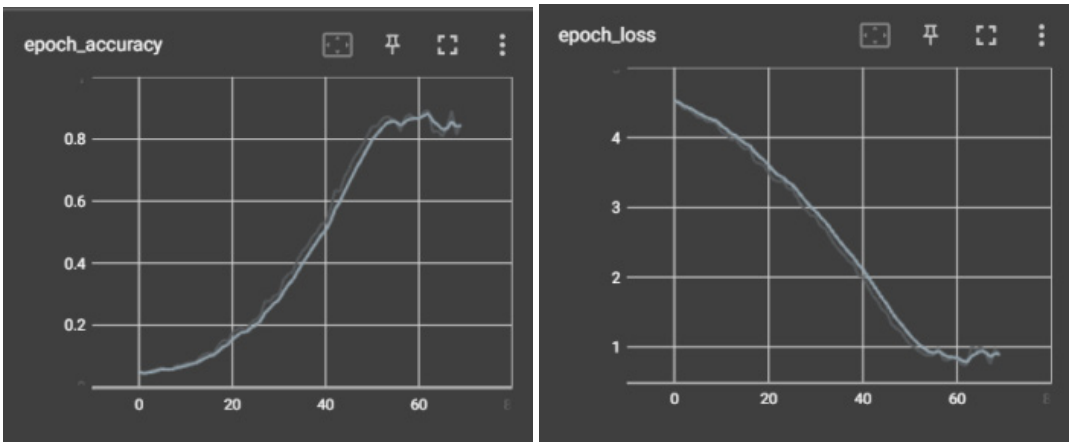


Fig. 11. Accuracy and Loss through the epochs

7. Conclusion

The MediaPipe sign language identification project is a potent example of how computer vision and machine learning technology may be combined to improve communication for those who use sign language. The project's success also represents a big step towards promoting a more open society and tearing down barriers to communication. Using unique machine learning models in conjunction with MediaPipe's hand posture estimation capabilities, the system is able to properly read and classify a wide variety of sign language movements. This discovery has enormous potential to close communication gaps and make interactions more inclusive and approachable for those with hearing loss. Projects such as these demonstrate the beneficial effects that creative solutions may have on the lives of people who use sign language as their major form of communication, even as technology advances. It is imperative that we keep improving these technologies as we go along to make sure that everyone on the planet can use them. We support more research and development in this area, embracing the collaborative nature of the open-source community and working towards the goal of making sign language recognition a global right rather than only a technological triumph. To increase the efficiency of the model the training dataset can be included with more data which can be trained further and will be able to detect the whole sentence instead of words and letters itself only.

References

[1] R.C. Gonzalez, R.E. Woods, Digital Image Processing, 2nd edn. (Prentice Hall, New Jersey, 2008), p. 693
[2] M.J.Z. Omar, M.H. Jaward, A review of hand gesture and sign language recognition techniques. Int. J. Mach. Learn. Cyber. 10, 131–153 (2019)
[3] C. Raghavachari and G. Shanmugha Sundaram, "Deep Learning Framework for Fingerspelling System using CNN," 2020 International Conference on Communication and Signal Processing (ICCSP), Chennai, India, 2020, pp. 469-473, doi: 10.1109/ICCSP48568.2020.9182155.
[4] A. M. Arjun, S. Sreehari and R. Nandakumar, "The Interplay Of Hand Gestures And Facial Expressions In Conveying Emotions A CNN–BASED APPROACH," 2020 Fourth International Conference on Computing Methodologies and Communication (ICCMC), Erode, India, 2020, pp. 833-837, doi: 10.1109/ICCMC48092.2020.ICCMC-000154.
[5] Sundar, B., and Bagyammal, T. (2022). American Sign Language Recognition for Alphabets Using MediaPipe and LSTM. Procedia Computer Science, 215, 642–651. https://doi.org/10.1016/j.procs.2022.12.066
[6] L. VB, S. KB, P. H, S. Abhishek and A. T, "An Empirical Analysis of CNN for American Sign Language Recognition," 2023 5th International Conference on Inventive Research in Computing Applications (ICIRCA), Coimbatore, India, 2023, pp. 421-428, doi: 10.1109/ICIRCA57980.2023.10220822.
[7] Simonyan, K. and Zisserman, A., 2014. Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.
[8] N. Aloysius and M. Geetha, "Understanding vision-based continuous sign language recognition", Multimed Tools Appl, vol. 79, pp. 22177-22209, 2020.
[9] S. Akshay, T. K. Mytravarun, N. Manohar and M. A. Pranav, "Satellite Image Classification for Detecting Unused Landscape using CNN," 2020 International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, 2020, pp. 215-222, doi: 10.1109/ICESC48915.2020.9155859.
[10] Mrinal Waila, "Object Detection vs. Image Classification vs. Keypoint Detection"
[11] Niklas Donges,"A Guide to Recurrent Neural Networks: Understanding RNN and LSTM Networks"
[12] https://www.turing.com/kb/recurrent-neural-networks-and-lstm
[13] S. Akarsh, S. Sriram, P. Poornachandran, V. K. Menon and K. P. Soman, "Deep Learning Framework for Domain Generation Algorithms Prediction Using Long Short-term Memory," 2019 5th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2019, pp. 666-671, doi: 10.1109/ICACCS.2019.8728544.
[14] Gabriel Guerin, "Sign Language Recognition - using MediaPipe and DTW"
[15] Jeremy Zhang, "Dynamic Time Warping"
[16] Ricardo Portilla, Brenner Heintz, "Understanding Dynamic Time Warping"